

TASK 5 Report - House Price Prediction using Linear Regression

Abstract

The objective of this project is to develop a machine learning model capable of predicting house prices using Linear Regression. The project demonstrates the complete machine learning workflow, including data preprocessing, exploratory data analysis, model training, evaluation, and visualization.

Introduction

House price prediction is a common regression problem in machine learning. Accurate prediction of property prices helps buyers, sellers, and real estate professionals make informed decisions. In this project, Linear Regression was used to model the relationship between housing features and house prices.

Dataset Description

The dataset contains various attributes related to residential properties, including:

- Area
- Number of Bedrooms
- Number of Bathrooms
- Parking Availability
- Main Road Access
- Guest Room Availability
- Furnishing Status
- House Price (Target Variable)

Methodology

Data Exploration

The dataset was examined to understand its structure, data types, and feature distributions.

Data Cleaning

- Checked for missing values.
- Converted categorical variables into numerical format.
- Prepared the dataset for machine learning.

Exploratory Data Analysis

A correlation heatmap was generated to identify relationships between features and the target variable.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio.

Linear Regression was applied using the Scikit-Learn library.

Model Evaluation

The model's performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Visualization

Visualizations were created to compare actual and predicted prices and to analyze feature importance.

Results

The Linear Regression model successfully predicted house prices based on the provided features. Evaluation metrics demonstrated the effectiveness of the model in estimating property values.

Conclusion

This project successfully implemented a Linear Regression model for house price prediction. The project provided practical experience in data preprocessing, exploratory data analysis, model training, performance evaluation, and result interpretation.

Future Improvements

- Implement Ridge and Lasso Regression.
- Use Ensemble Models such as Random Forest Regressor.
- Perform advanced feature engineering.
- Deploy the model as a web application.

Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Scikit-Learn
- Jupyter Notebook

Outcome

The project enhanced understanding of regression analysis, machine learning workflows, and predictive modeling techniques.